Instructions:
1. Run task_0a.py to generate the vector database if not already generated.
2. Press Run All (or restart kernel and run all cells).
3. You will be prompted to provide input values.

– Task 0b: Implement a program which, given (a) an (even or odd numbered) imageID or an image file, (b) a user selected feature space, and (c) positive integer k, identifies and visualizes the most similar k images, along with
their scores, under the selected feature space.

In [ ]:

```
IMAGE_INPUT = input(
"""
Provide one of the following:
1. An Image ID in the Caltech101 dataset in range [0, 8676].
2. The name of an image file in /Code/input/ directory (eg: image.jpg).
"""
)

FEATURE_SPACE = input("Provide a feature space [color, hog, avgpool, layer3,

K = int(input("Enter K, the number of similar images K to find for the selec
```

In [ ]:

```
# Find the k most similar images for a given query image for the given featu

from utils.query_input_processor import get_query_image

image = get_query_image(IMAGE_INPUT)

print("Input image: ", IMAGE_INPUT)
display(image)
```

```
Input image:  image.jpg
```

In [ ]:
```
if FEATURE_SPACE == 'color':
    from feature_models.color_moments import ColorMomentsExtractor

    vector = ColorMomentsExtractor(image).get_color_vector()

elif FEATURE_SPACE == 'hog':
    from feature_models.hog import HOGExtractor

    vector = HOGExtractor(image).get_hog_vector()

else:
    # ResNet feature model.
    from feature_models.resnet import ResNetExtractor
    resnet = ResNetExtractor(image)

    if FEATURE_SPACE == 'avgpool':
        vector = resnet.get_avgpool_vector()

    elif FEATURE_SPACE == 'layer3':
        vector = resnet.get_layer3_vector()

    else:
        # fc feature space.
        vector = resnet.get_fc_vector()
```

In [ ]:
```
# Load feature space vector database into memory.

from utils.dataset_utils import initialize_dataset
dataset = initialize_dataset()

from utils.database_utils import retrieve
feature_vectors = retrieve(f'{FEATURE_SPACE}.pt')

from utils.distance_utils import get_distance_fn, top_k_distance_ranker
top_k = top_k_distance_ranker(K, vector, feature_vectors, get_distance_fn(FE

print("\nK =", K, " most similar images by ", FEATURE_SPACE, " feature space

for i in range(len(top_k)):

    distance, image_id, label = top_k[i]

    print(str(i + 1) + ".", "\tImage ID: ", image_id, "\t\tLabel: ", label,
    display(dataset[image_id][0])
```

```
Downloading dataset if not present.
Files already downloaded and verified

K = 10  most similar images by  fc  feature space using distance:  cosine
1.      Image ID:  2254                 Label:  airplanes               Dist
ance:  0.11802619695663452
```

```
2.      Image ID:  2308                 Label:  airplanes               Dist
ance:  0.1238332986831665
```

3. Image ID: 2226 Label: airplanes Distance: 0.12936222553253174

4. Image ID: 2670 Label: airplanes Distance: 0.1307840347290039

5. Image ID: 2484 Label: airplanes Distance: 0.14001035690307617

6. Image ID: 2500 Label: airplanes Dist
ance: 0.1507652997970581

7. Image ID: 2490 Label: airplanes Dist
ance: 0.1560509204864502

8. Image ID: 2194 Label: airplanes Dist
ance: 0.15802335739135742

9. Image ID: 2134 Label: airplanes Distance: 0.16020268201828003

10. Image ID: 2492 Label: airplanes Distance: 0.164375901222229